



## Python Programming - 2301CS404

### Lab - 4

Roll No.:310

Name:Ronit Bhadania

#### 01) WAP to print 1 to 10.

```
In [1]: for i in range(1,11) :  
        print(i,end=" ")
```

1 2 3 4 5 6 7 8 9 10

#### 02) WAP to print 1 to n.

```
In [2]: n = int(input("Enter the number:"))  
        for i in range(1,n+1) :  
            print(i,end=" ")
```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

#### 03) WAP to print odd numbers between 1 to n.

```
In [4]: n = int(input("Enter the number:"))  
        for i in range(1,n+1,2) :  
            print(i,end=" ")
```

1 3 5 7 9 11 13 15 17 19 21 23 25 27 29

#### 04) WAP to print numbers between two given numbers which is divisible by 2 but not divisible by 3.

```
In [39]: a = int(input("Enter first number:"))  
         b = int(input("Enter second number:"))  
         for i in range(a,b) :  
             if(i%2==0 and i%3!=0) :  
                 print(i,end=" ")
```

2 4 8

**05) WAP to print sum of 1 to n numbers.**

```
In [8]: n = int(input("Enter the number:"))
sum=0
for i in range(1,n+1) :
    sum = sum + i
print(f"Sum={sum}")
```

Sum=55

**06) WAP to print sum of series 1 + 4 + 9 + 16 + 25 + 36 + ...n.**

```
In [10]: n = int(input("Enter the number:"))
sum=0
for i in range(1,n+1) :
    sum = sum + i*i
print(f"Sum={sum}")
```

Sum=385

**07) WAP to print sum of series 1 – 2 + 3 – 4 + 5 – 6 + 7 ... n.**

```
In [11]: n = int(input("Enter the number:"))
sum=0
for i in range(1,n+1) :
    if(i%2==0) :
        sum = sum - i
    else :
        sum = sum + i
print(f"Sum={sum}")
```

Sum=4

**08) WAP to print Multiplication Table of the given number.**

```
In [13]: n = int(input("Enter the number:"))
for i in range(1,11) :
    print(f"{n}*{i}={n*i}")
```

```
19*1=19
19*2=38
19*3=57
19*4=76
19*5=95
19*6=114
19*7=133
19*8=152
19*9=171
19*10=190
```

**09) WAP to find Factorial of the given number.**

```
In [14]: n = int(input("Enter the number:"))
fact=1
```

```
for i in range(1,n+1) :  
    fact=fact*i  
print(f"Factorial={fact}")
```

Factorial=120

## 10) WAP to print GCD of given two numbers.

```
In [15]: a = int(input("Enter first number:"))  
b = int(input("Enter second number:"))  
while(b!=0) :  
    temp = b  
    b=a%b  
    a=temp  
print(f"GCD={a}")
```

GCD=3

## 11) WAP to find Factors of the given number.

```
In [16]: n = int(input("Enter the number:"))  
for i in range(1,n+1) :  
    if(n%i==0) :  
        print(i,end=" ")
```

1 2 3 4 6 12

## 12) WAP to find whether the given number is Prime or not.

```
In [18]: n = int(input("Enter the number:"))  
count=0  
for i in range(1,n+1) :  
    if(n%i==0) :  
        count=count+1  
if(count==2) :  
    print("Prime Number")  
else :  
    print("Not Prime Number")
```

Prime Number

## 13) WAP to print sum of digits of given number.

```
In [19]: n = int(input("Enter the number:"))  
sum=0  
while(n!=0) :  
    rem=n%10  
    sum = sum + rem  
    n=n//10  
print(f"Sum={sum}")
```

Sum=6

## 14) WAP to check whether the given number is Palindrome or not.

```
In [23]: n = int(input("Enter the number:"))
rev=0;num=n
while(num!=0) :
    rem=num%10
    rev=rev*10 + rem
    num=num//10
if(rev==n) :
    print("Palindrome Number")
else :
    print("Not Palindrome Number")
```

Palindrome Number

## 15) WAP to check whether the given number is an Armstrong Number or not.

```
In [28]: import math
n = int(input("Enter the number:"))
sum=0;num=n;digit=0;x=n
while(x!=0) :
    x=x//10
    digit+=1
while(num!=0) :
    rem=num%10
    sum = sum + math.pow(rem,digit)
    num=num//10
if(sum==n) :
    print("Armstrong Number")
else :
    print("Not Armstrong Number")
```

Armstrong Number

## 16) WAP to print all the perfect numbers between 1 to n.

```
In [30]: n = int(input("Enter the number:"))
for i in range(1,n+1) :
    sum=0
    for j in range(1,i) :
        if(i%j==0) :
            sum = sum + j
    if(sum==i) :
        print(i,end=" ")
```

6 28